

WearSway: Wearable Device to Reproduce Tactile Stimuli of Strong Wind through Swaying Clothes

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Figure 1: (A) WearSway device. (B) WearSway presents the sensation of strong wind by swaying clothes. (C, D) Examples of virtual scenes to experience strong wind using WearSway.

ABSTRACT

Wind displays reproduce cutaneous stimuli of wind by blowing actual airflow to make virtual content more immersive. However, reproducing the sensation of a strong wind across the whole body, as might be used in educational or entertainment scenarios, typically requires large and cumbersome wind fans. WearSway is a compact and lightweight wearable haptic device that simulates the swaying of clothes in the wind to express the strong wind without large equipment. The device achieves this by using motors and strings to sway the clothing and a small fan to provide wind sensation. While most existing wind displays stimulate the skin directly, WearSway introduces a novel approach by leveraging the movement of clothing to deliver a strong wind simulation.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Wind display, Wearable Device, Haptics, Virtual Reality

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1 INTRODUCTION

Wind displays, devices that reproduce cutaneous stimuli of wind, have been utilized to enable immersive virtual content [da Silveira et al. 2023]. There are instances where we need to reproduce strong wind blowing to the whole body, such as disaster prevention education focusing on tornadoes [Mitsuhara et al. 2020] or entertainment utilizing flight experience [Rheiner 2014]. However, a large and powerful fan is needed to blow strong wind to the whole body. Such a fan takes up space, consumes much energy, and entails the risk of blowing objects away with an exceedingly strong wind. Thus, we propose WearSway, a wearable device that makes clothes sway as a method of providing the tactile stimuli of strong winds without large equipment (Figure 1).

When situated in a strong gale, we experience the phenomenon of our clothes fluttering vigorously. WearSway reproduces such fluttering of the clothes using strings and motors and can be combined with small fans. While this method does not reproduce all the sensory stimuli of wind, it is considered to help users perceive a portion of the effects of strong winds. Since WearSway is compact and lightweight, it can be easily worn and unintrusive. Overall, WearSway shows a new possibility for wind displays by utilizing clothes that have rarely been employed to generate wind sensations.

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2 RELATED WORK

While most existing haptic displays stimulate the skin directly, some designs generate stimuli indirectly by moving clothes. FiveStar VR reproduced a tactile stimulus of water droplets from a waterfall by vibration actuators attached to rubber sheets positioned on the shoulders [Shimizu et al. 2018]. Oishi et al. proposed HapPull, which induces a sense of acceleration while moving by pulling the user's body forward and backward using wires extended from frames surrounding them [Oishi et al. 2018]. Ito et al. recreated the sensation of being swayed by strong wind in their virtual reality (VR) installation of experiencing firefighting by inducing the vibration of clothing through strings connected to a motor [Ito et al. 2019]. Although the devices using strings have the advantage that they can move clothes without any direct contact with rigid parts, they restrained the user's movement as a result of the strings being attached to a fixed base. Therefore, we designed WearSway to make motors that drive the strings wearable on the user's body, making them less intrusive to their movements.

3 IMPLEMENTATION AND EVALUATION

WearSway comprises a cloth unit that can be worn over the arm and an actuator unit equipped with a motor to vibrate the cloth using strings (Figure 1 (A)). We utilized Hapbeat [Yamazaki et al. 2017] for the actuator part, a device that produces vibrotactile stimuli using strings. Hapbeat is capable of vibrating the string connected to the motor with a frequency range of 0 to 600 Hz. By attaching the actuator unit to the arm using a belt and connecting the end of the string to the cloth, we can make the cloth sway with the vibrating string (Figure 1 (B)). The volume during operation is about 60 dB.

To simulate the swaying of clothes in the wind, we measured the movement of the cloth unit when exposed to airflow from a hairdryer (Figure 2 (A)). We positioned the cloth unit 400 mm away from a hairdryer (Dyson Supersonic) and directed the wind with a velocity of 15 m/s. The movement of the cloth was measured using a laser displacement meter (KEYENCE LK-G5000V). As a result, we observed the peak vibration around 13 Hz (Figure 2 (B)). Therefore, we input a 13 Hz sinusoidal wave to WearSway and confirmed that the cloth sways with a fundamental frequency of 13 Hz using the laser displacement meter (Figure 2 (C)).

4 EXPERIENCE

WearSway is considered capable of producing not only a sense of strong wind for diverse virtual content but also fictional haptic effects. We have designed three demonstrations to showcase the applicability of WearSway from education to entertainment. Attendees will be able to experience the three demonstrations in sequence, each containing a VR video or an interactive mini-game. We also incorporate actual wind using a small fan so that users can experience the perceived wind velocity enhanced by the device.

Violent Storm Experience. In this demonstration, WearSway is employed to enhance the experience of a simulated storm, commonly presented in disaster prevention education (Figure 1 (C)). Users can genuinely experience the power of strong wind through the sensation of their clothes swaying, in addition to the immersive video and sound of a heavy rainstorm.

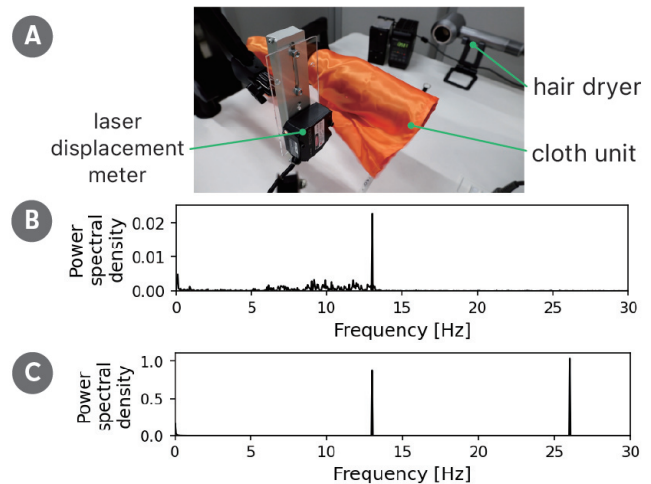


Figure 2: Measurement results of swaying clothes. (A) Apparatus for measurement. (B) Swaying frequency of cloth in the actual wind. (C) Swaying frequency of WearSway.

Paramotor Flight. WearSway is utilized to simulate the haptic stimulus of wind in a flying experience. Users take control of a paramotor and navigate through a virtual environment, experiencing the airflow from the front through WearSway (Figure 1 (D)). Furthermore, we adjust the vibration intensities between the user's left and right arms when the users make a turn to make the experience more immersive.

Magical Battle. As an example of fictional haptic effects, the magical battle demonstration offers an immersive experience of magical energy through WearSway. When users launch magical attacks or defend against them by moving their arms, WearSway vibrates to simulate the haptic feedback of magic.

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REFERENCES

- Aleph Campos da Silveira et al. 2023. Thermal and wind devices for multisensory human-computer interaction: an overview. *Multimedia Tools and Applications* (07 Mar 2023). <https://doi.org/10.1007/s11042-023-14672-y>
- Koichiro Ito et al. 2019. Virtual Firefighting -ENBU-. <https://ivrc.net/2019/en/projects-en/>. 27th International collegiate Virtual Reality Contest (IVRC).
- Hiroyuki Mitsuhashi et al. 2020. Comparative Experiments on Simulated Tornado Experience via Virtual Reality and Augmented Reality. *The Journal of Information and Systems in Education* 19, 1 (2020), 21–31. <https://doi.org/10.12937/ejsise.19.21>
- Erika Oishi et al. 2018. HapPull: Enhancement of Self-motion by Pulling Clothes. In *Advances in Computer Entertainment Technology (ACE 2017)*, Adrian David Cheok, Masahiko Inami, and Teresa Romão (Eds.). Springer International Publishing, Cham, 261–271. https://doi.org/10.1007/978-3-319-76270-8_18
- Max Rheiner. 2014. Birdly an Attempt to Fly. In *ACM SIGGRAPH 2014 Emerging Technologies* (Vancouver, Canada) (SIGGRAPH '14). ACM, Article 3, 1 pages. <https://doi.org/10.1145/2614066.2614101>
- Koichi Shimizu et al. 2018. FiveStar VR: Shareable Travel Experience through Multisensory Stimulation to the Whole Body. In *SIGGRAPH Asia 2018 Virtual & Augmented Reality* (Tokyo, Japan) (SA '18). ACM, Article 2, 2 pages. <https://doi.org/10.1145/3275495.3275502>
- Yusuke Yamazaki et al. 2017. Hapbeat: Single DOF Wide Range Wearable Haptic Display. In *ACM SIGGRAPH 2017 Emerging Technologies* (Los Angeles, California) (SIGGRAPH '17). ACM, Article 12, 2 pages. <https://doi.org/10.1145/3084822.3084843>